DEPARMENT OF INFORMATION TECHNOLOGY
FACULTY OF ENGINEERING AND TECHNOLOGY

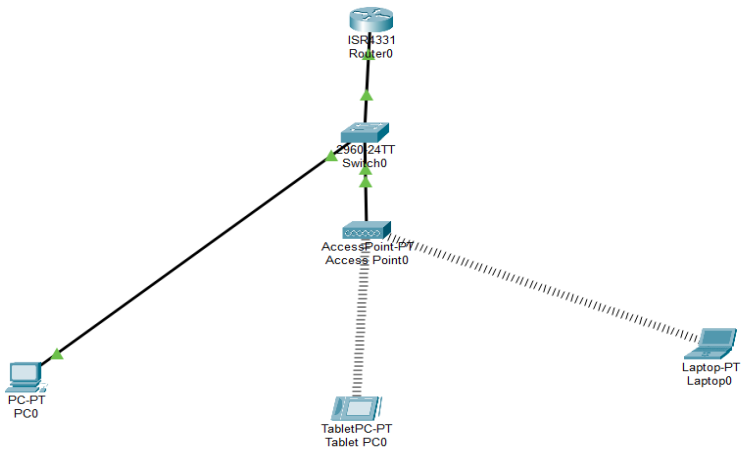
SE25-425 P-II
(MORNING/EVENING)
COMPUTER NETWORKS

LAB#5

TASK: IMPLEMENTATION OF WIRELESS LAN

Name: _____ Roll No:_____

Score: _____ Signature of the Lab Tutor _____ Date: _____



Wireless LAN is defined as IEEE 802.11 standard. The Wireless LAN can be defined using two architectures as: adhoc or infrastructure based. In the infrastructure based networks, the access point is used to connect the different devices. In the adhoc network there is no such requirement.

Objective

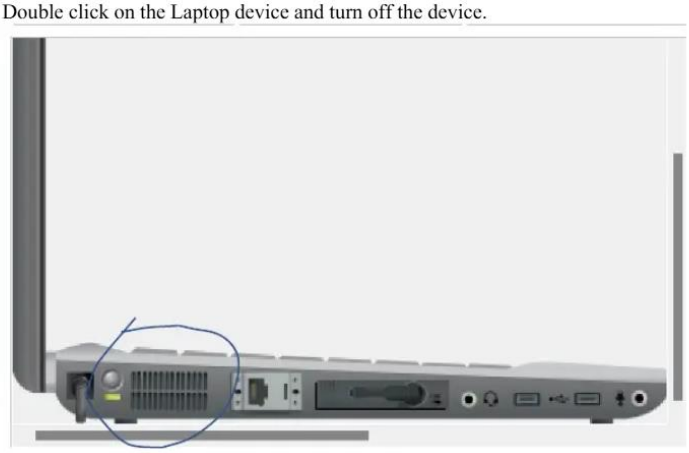
We want to create a wireless LAN, which consist of a PC, a laptop and wireless tablet as shown in the Figure.

Steps

The steps to create the network is as follows:

1. Add a switch device by selecting from the Networking Devices menu.
2. Add a router device by selecting from the Networking Devices menu.
3. Add a PC device by selecting from the End Devices menu.
4. Add a laptop device by selecting from the End Devices menu.
5. Add a wireless tablet device by selecting from the End Devices menu.
6. Add an Access Point device by selecting from the sub menu of Wireless Devices from the Networking Devices menu.
7. The wireless tablet device automatically gets connected to the access point.
8. To connect the Laptop device with the access point follow the following steps:

a. Double click on the Laptop device and turn off the device.



The yellow light turns off which shows the device is in off state.

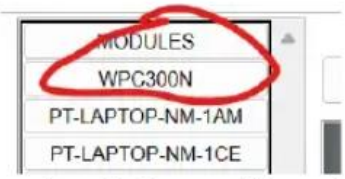
b. In the next select and remove the existing interface from the laptop and add it to the interface list.



After removing the existing port, the laptop looks like this



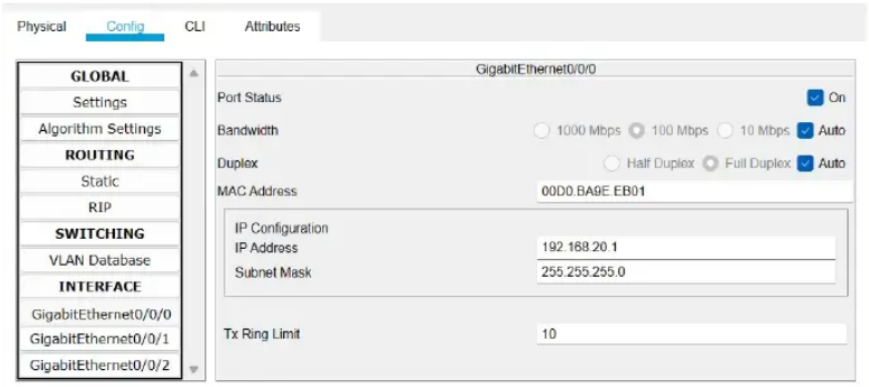
c. Add the wireless interface to the laptop.WPC300N is the wireless interface.



After adding the wireless interface the laptop will now look like this.

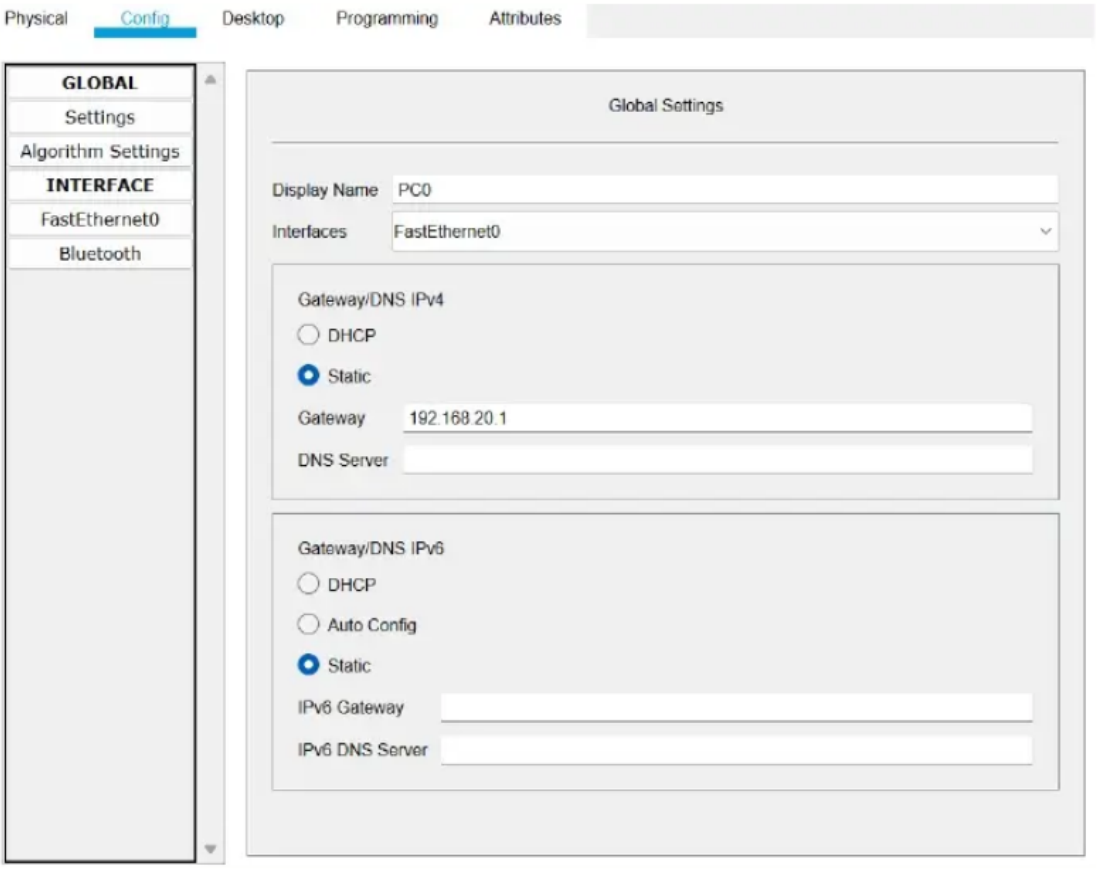
d. Switch on the laptop.

9. Assign the IP address to the router by selecting the appropriate interface GigabitEthernet 0/0/0 from the Configure menu as shown. Switch on the Port Status.



The router will act as the default gateway for all the devices connected in the network.

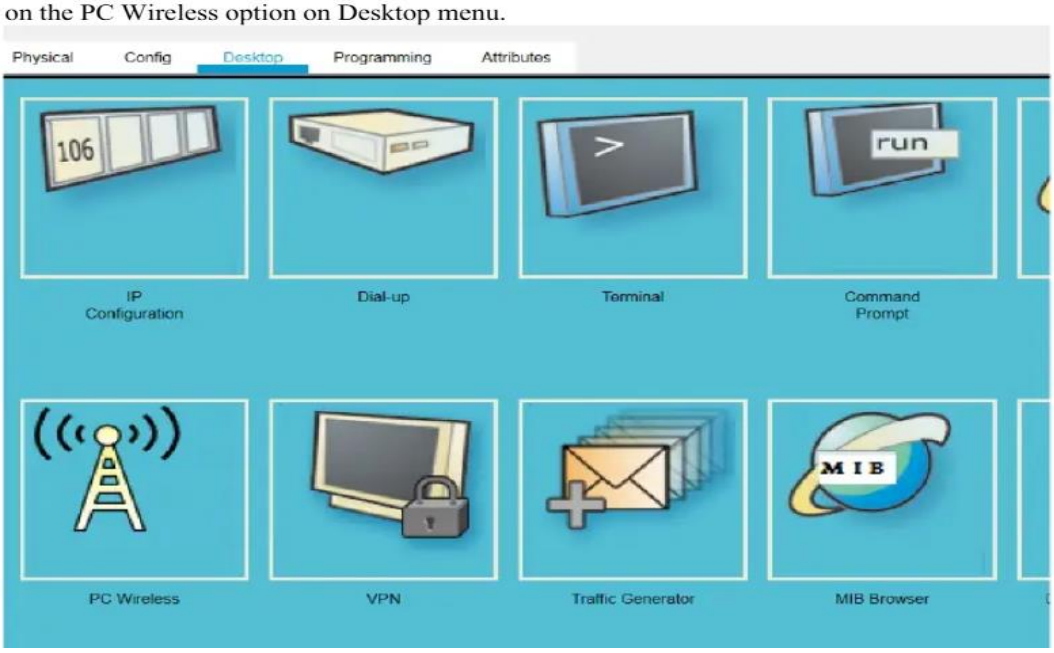
10. Assign the IP address to the PC by selecting the config option after double clicking on the PC icon as shown:

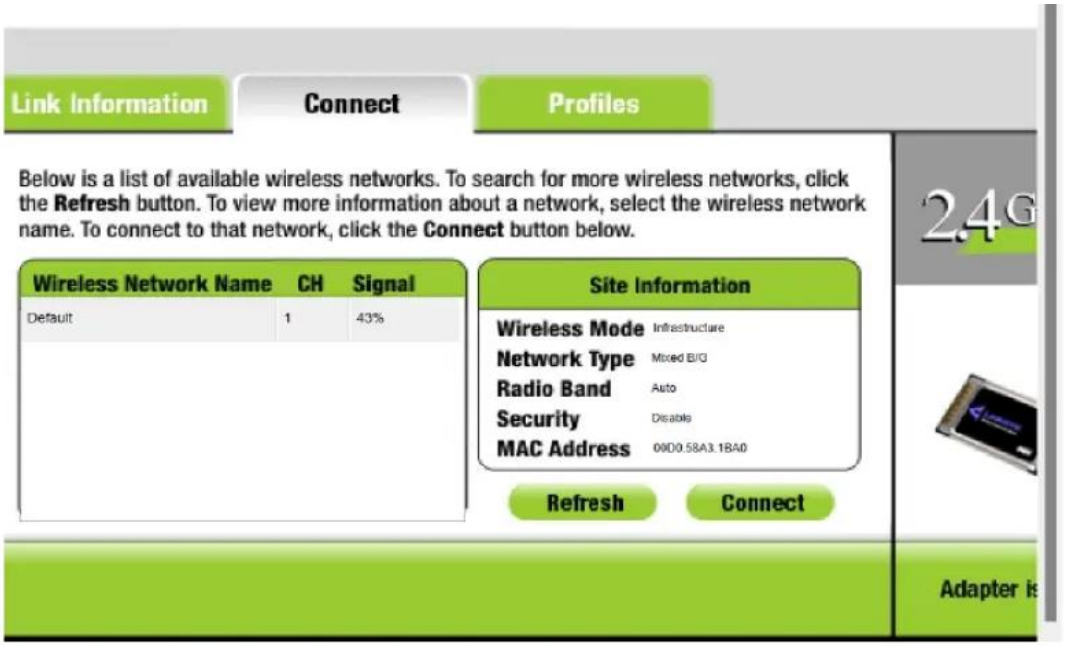


Similarly, assign the IP addresses to all the remaining devices.

11. The access point can be configured to enable any new device to connect to it using password. Click on the Config and enable the WEP mode and assign the key to it.

12. The laptop can be manually connected to the access point or any network by clicking on the PC Wireless option on Desktop menu.





Click on the refresh option to see the available network options and connect using the connect option.

13. The network communications can be verified by transmitting the PDU between any of the two devices.

Review Questions:

1. How does WiFi work?

Answer:

2. What is the main function of a WiFi router?
   Answer:

3. How do you set up a WiFi network at home?
   Answer:

4. How can you secure a WiFi network?
   Answer:

Note: Submit the handout before the next Lab